

Three Roots for Leibniz's Contribution to the Computational Conception of Reason

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Abstract: *The aim of this paper is to show that Leibniz's contribution to what we may call today the computational conception of reason must be inscribed in three different roots: 1) the combinatory tradition coming from the XIII century, 2) XVII century attempts at constructing an artificial universal and philosophical language, and 3) the Hobbesian conception of reason.*

1. Lull and Leibniz

Leibniz first great inspiration is the old *Ars Magna* by Ramón Lull (1232-1315). Lull's central idea is that it would be possible, by the combination of a set of simple terms, to establish all possible propositions and thus to discover all possible statements and demonstrate all possible truths to which human knowledge can aspire.

For the accomplishment of this project, Lull a) establishes a set of categories, organized in six great classes of nine categories each; b) defines the syntactic rules of the combinatorial elements; c) conceives a complex system of technical procedures of automatic application capable of effecting the combination of these elements. His aim is to make it possible to determine, automatically, all the possible subjects of a given attribute as well as the conclusion and the middle term of all incompletely known syllogisms¹. Thus, starting on the basis of categories of universal application and operating with a system of symbolic notations and combinatorial diagrams, Lull establishes the principles of a synthetic and inventive procedure which, unlike the demonstrative logics of Aristotle, would not be limited to the analysis of truths already known but would make possible ways for the discovery of new truths.

¹ Among the complex set of signs, tables and figures constructed by Lull, there is a set of material circles, rotating in concentric movement of superposition in order to allow the combination of the symbols marked in their limits, cf. Carreras y Artau (1939: 345-455) e também Couturat «*Sur L'Ars Magna de Raymond Lulle et Athanase Kircher*» (1981: 541/3).

Lull's *Ars Magna*, which exerted a deep influence in the renaissance and modern times², is thus one of the most far-reaching and prestigious proposals for the mechanization of logical operations.

Leibniz was quite aware of Lull's *Ars*. In fact, since his first writings, namely *De Arte Combinatória* (1660), Leibniz criticizes a) the arbitrariness of Lullian signs; b) his set of categories; c) the methodological aspects of Lull's *Ars* which Leibniz considers insufficient and rudimentary. These are decisive critiques. Leibniz cannot accept the alphabet of the Lull's *Ars*. He cannot admit his list of categories which he considers as «vagues», arbitrarily chosen and organized in classes of 9 categories each, a number artificially chosen purely for reasons of symmetry³. Further, he does not agree with the mechanical combinatorial procedures proposed by Lull. Instead, Leibniz points to the calculatory processes of mathematical analysis (binary, ternary, quaternary groupings, exchanges, substitutions, equivalencies, etc.). On the basis of the analogy between the analysis of the concepts, its associative structure, and the decomposition of a whole number into prime factors, Leibniz proposes, in the *De Arte*, to assign a whole number to each primitive idea, each complex idea therefore being represented by the product that would constitute its definition.

However, behind these criticisms, Leibniz recognizes the epistemological value of Lull's project on two important points: a) the status of combinatory as the foundation for the *ars inveniendi*; b) the possibility of applying the same logical procedures to the totality of knowledge, including new knowledge. That is to say, Leibniz accepts Lull's idea that the knowledge of the primitive ideas and of the laws of their combination would allow us to find all possible predicates of a given subject and vice versa, that is, not only formulate all the truths already known but also to invent/to discover new truths⁴. By taking mathematics as a model, Leibniz conceives the project of arithmetization of all the mechanisms of thought, of submitting the totality of human intellectual activity to calculatory processes able not only to demonstrate all known propositions but also to discover all new propositions, that is, to reduce rational activity to a calculus. However, in my view, Leibniz's greatest difference with respect to Lull concerns the kind of symbolic system able to operate within the calculus. Anticipating his fundamental methodological experiment - the discovery of the algorithm of the differential calculation as a true *organon* of mathematical research⁵ - Leibniz is fully aware of the value of the sign as a not merely representative but also prospective and heuristic tool. But for that, Leibniz claims that the signs must not be, as in Lull, arbitrarily chosen. That is, Leibniz's project is not to be reduced to the constitution of a mere formal language, a system of universal signs that would allow the logical treatment of science and would supply a set of simple, necessary and rigorous symbols to express all actual and possible knowledge. Leibniz does not fall into the illusion – in some way present in Lull's project – according to which the automatic functioning of a set of operative rules can

² Namely the encyclopaedic lullism of Johann Heinrich Alsted (1588-1638) or Sebastián Izquierdo (1601-1681), and the major work of Athanasius Kircher, *Ars Magna sciendi seu porta Scientiarum*, published in 1669, which claims to be the development of Lull's *Ars Magna*

³ Cf. for instance (GP 4: 63).

⁴ Cf. *De Arte Combinatoria*, §§ 64/79 (GP 4: 64-8).

⁵ As Cassirer stress (1923-29.1: 76), Leibniz would have applied his methodological experience with the algorithm of the differential calculation to the construction of the universal language.

allow the development of science. For Leibniz science cannot be reduced to a merely formal, well-made language. That is why heuristics in Leibniz also pass for the requirement of semantics.

2. XVII century projects of philosophical language and Leibniz

Let us now come to the XVII century and to the several attempts then made for the construction of an universal philosophical language. The XVII century was a time of an intense research on the intimate nature of human languages. The reasons are many and diverse and I will not analyze them here⁶. Let me just point out that the main question concerns the role which language performs in the process of knowledge: Does language help to promote knowledge? Or, conversely, does it prevent its progress? Is language a disturbing factor or a necessary element for the acquisition of knowledge? A mere means for communicating knowledge or an essential *medium* for constituting knowledge?

Two main positions can be identified: a) a critical position, which attributes to natural languages mere communicative functions and moreover emphasizes their insufficiencies and disturbing effects and b) a positive position which, although recognizing some limits and imperfection in human languages, nevertheless stresses its constitutive character, that is, its decisive role as a necessary cognitive device. Both positions were defended by great thinkers. In the first critical position we find Bacon, Locke, Descartes, Arnauld, Malabranche and, in general, all those who look for the construction of new artificial languages – from the baroque English pasigraphers to the constructors of philosophical languages (from Lodwick to Dalgano, from Seth Ward to Wilkins). In the second positive position, we can only find two names in modern times: Thomas Hobbes and some years later Leibniz.

Now, what I would like to stress is the fact that a) both of these two conflicting positions were relevant for the development of a computational conception of reason; and b) that Leibniz was the only one who fully understood the possibility of making those opposite positions converge in a fully developed conception of linguistic reason.

The critical position comes mainly from the awareness of the inadequateness of natural languages for representing the universe that modern science is providing as well as the logical categories of thought that are the basis for that scientific progress.

Bacon (1561-1626) is the great inspiration. He is the first to design the project of constructing a universal language which would represent, «neither letters nor words, but things and notions» (Bacon, 1605: 6.1.439). The idea was to build a language on the basis of a set of (real) characters representing not the sounds or voices of any natural language but the things or notions. This language could thus be read and understood by all men in all different languages. As Bacon writes: «Any book written in characters of this kind can be read of by each nation in their own language» (Bacon, 1605: 6.1, 439). Bacon also points towards the construction of a philosophical grammar, a set of rules common to all vulgar natural tongues. In his words : «If some one well seen in a great number of tongues, learned as well as vulgar, would handle the various properties of languages, showing in what points each excelled, in what it failed. For so not only many languages be enriched by

⁶ For further developments on these issues, see Pombo (1987).

mutual exchanges, but the several beauties of each may be combined <...> into a most beautiful image and excellent model of speech itself, for the right expressing of the meanings of the mind» (Bacon, 1605: 6.1.421-422).

The programmatic suggestions of Francis Bacon, presented mostly in *The Advancement of Learning* (1605), are at the root of numerous projects of construction of a universal artificial language which emerged in the XVII century. Their aim is the construction of a universal communicative tool. These projects, which gained great popularity in the XVII century, are clear practical endeavours. Facing mostly the needs created by the development of international contacts (economical, political, scientific and religious), their aim is the construction of a universal language, unique, perfectly regular, able to substitute Latin, to overcome the imperfections and diversity of natural languages and to allow adequate and full communication between men. The idea pursued is that of pasigraphy, a word defined by the XIX century Larousse as «the art of writing or printing in the only tongue that one knows in order to be able to be read and to be understood, without necessity of translation, in any another tongue that one does not know”. The goal is effective communication among peoples of different tongues by a system of written notations, independent of pronunciation, by means of which it would be possible to establish direct correspondences between the different words of different languages. Any individual, independently of their language, would thus be capable of reading and understanding any codified text.

Of the many works undertaken in the XVII century, we must refer those of the English school of Baroque pasygraphs, Cave Beck, *The Universal Character, by which all the Nations in the World may Understand one Another's Conceptions* (1657), Henry Edmundson, *The Natural Languages in a Vocabulary Contrived and built upon Analogy* (1658), Edward Somerset, *A Century of the Names and Scantlings of such Inventions as at Present I can call to Mind to have Tried and Perfected* (1663). Above all, must be mentioned the works of the German Johan J. Becher, *Character pro Notitia Linguarum Universali* (1661) as well as of Athanasius Kircher, *Polygraphia Nova et Universalis ex Combinatoria Arte Delecta* (1663). Starting out from the assignment of a number to each word of the Latin dictionary and the organization of dictionaries of diverse languages in which each word was to be followed by the same number as attributed to its Latin equivalent, Becher intended to allow the translation, in any language, of a text written solely with numerical signs. The reader would only need to have access to the key-dictionary organized for their own language. More elaborated, Kircher's project consists in the construction of two polyglot dictionaries (Latin, Italian, French and German), each one organized in view of its future communicative role as source and target⁷. In both cases, we are faced with written systems of codification, totally artificial and arbitrary, of the sort we today would call special and technical languages (graphical semiotic systems limited to domains such as codes of telegraphic or maritime signals) rather than true

⁷In the dictionary designed for sending the message, the words are ordered alphabetically and each one corresponds to the number of its page (in Arab numbers). In the dictionary designed for the reception of the message, the synonyms of the five languages are displayed in columns according to the alphabetical order of the Latin words⁷. We meet here a project which supposes a sketch of classification of concepts, that is, a philosophical project. For Leibniz's position with regard to Becher and Kircher, cf. (GP 4: 72), (C: 536) and his *letter to Oldenburg July 1670* (GP 7: 5).

philosophical languages. However, their authors, directly inspired by Bacon's programmatic proposals, looked for the construction of artificial languages able to avoid the ambiguity and irregularity of natural tongues, thus allowing the development of commerce, of knowledge and of true religion. As Cave Beck writes in the preface to *The Universal Character* (1657), such a universal language would be «a singular means of propagating all sorts of learning and true religion in the world».

Philosophical projects are more ambitious. Beyond merely communicational goals, their aim is to construct a symbolic system able adequately to express thought and its articulations and to supply a rigorous symbolic system capable of translating all the actual and possible knowledge, that is, to fulfil an essentially cognitive role. The idea is that it is possible to reduce all our ideas and notions to a small number of basic concepts, simple or primitive ideas. Universal language would have then to consist in the assigning of a character, not to each of the possible ideas as Bacon argued, but to each of the elementary ideas, and in the definition of a grammar (fixed rules of combination of characters). The inspiration is now cartesian. The primary objective is to find the small number of simple ideas to which all other can be reduced and, by the combination of an equally limited number of characters, each one corresponding to one of those simple ideas, to construct the whole system of knowledge. That is why, prior to the choice of the characteristic signs, all philosophical projects begun by making an elaborated logical and semantical classification of concepts.

In a celebrated *Letter to Mersenne*, 20th November 1629 (AT 1:76-82), Descartes had in fact stated a very relevant thesis about the possibility of a universal language. If the totality of arithmetic may be constructed upon a small number of axioms, in a similar way, it would have to be possible to exhaustively symbolize the totality of the thoughts on the basis of a limited number of linguistic signs. As he writes : «Je trouve qu'on pourroit ajouter à ceci une invention, tant pour composer les mots primitifs de cette langue que pour leurs caractères en sorte qu'elle pourroit être enseignée en fort peu de temps, et ce par le moyen de l'ordre, c'est-à-dire établissant un ordre entre toutes les pensées qui peuvent entrer en l'esprit humain de même qu'il y en a un naturellement établi entre les nombres; et comme on peut apprendre an un jour à nommer toutes les nombres jusques à l'infini et à les écrire en une langue inconnue, qui sont toutes fois une infinité de mots différents, qu'on pût faire le même de toutes les autres mots nécessaires pour exprimer toutes les autres choses qui tombent en l'esprit des hommes» (AT1: 80-81).

The paradigm is thus mathematics as a symbolic system of universal application and, simultaneously, underlying the order of the world and the structure of the creation. That is to say, mathematics is not only a deductive system grounded on the evidence of its axioms or a rigorous method of demonstration, but a universal language which may represent the world because it is isomorphic with its structure, because the world is, in itself, mathematically ordered and structured. In this sense, it is important to remember that the mathematics which is the basis of modern science is, in all cases, warranted by the Pythagorean conception of the mathematical structure of creation (see the cases of Galileo, Descartes, Kepler, Leibniz or Newton). So, the ideal of a universal philosophical language is an extension and a deepening of mathematical procedures able to operate on the basis of universal categories able to

develop and uncover the integral system not only of knowledge but also of world and things.

Of the many philosophical languages proposed in the XVII century, the first seems to have been that of Mersenne who, in a letter to Fabri de Peiresc dated of 1636 or 37⁸, claims to have finished the construction of a philosophical language, surely of cartesian inspiration.⁹ In England too, we ought to mention names such as the mathematician and astronomer Seth Ward, *Vindiciae Academicarum* (1654). In terms close to those used by Descartes, they both point to the decomposition or analysis of concepts into primitive terms¹⁰. Also in England, Georges Dalgarno and John Wilkins realized the most relevant and complete philosophical languages. Dalgarno publishes *Ars Signorum, vulgo Character Universalis et Lingua Philosophica* (1661) and Wilkins, bishop of Chester and one of the most reputed scholars of the period, publishes in 1668, with the support of Royal Society, *An Essay towards a Real Character and a Philosophical Language, with an Alphabetical Dictionary*. On the basis of a hierarchical organization of concepts in classes and sub-classes, Dalgarno points to a logical symbolic system of letters able to express the normal articulation of thought¹¹. Similarly, Wilkins starts out from a classification of concepts which he organizes in a complex system of *Summa Genera, Differentia and Species*. Those categories, with which Wilkins aims to cover the totality of scientific knowledge,¹² are represented by a *real character*, that is, not by letters as in Dalgarno, but by written characters of ideographic nature¹³.

Leibniz knew well all this philosophical projects. He praised above all Wilkins's project. «J'ai considéré avec attention le grand ouvrage du Caractere reel et Langage Philosophique de Mons. Wilkins; je trouve qu'il y a mis une infinité de belles choses, et nous n'avons jamais eu une Table des predicateurs plus accomplie» (*Letter to Burnet, 24th August 1697, GP 3: 216*). However, Leibniz does not avoid criticism. His main concern is the insufficient analysis which, in Leibniz opinion, underlies Dalgarno and Wilkins proposal. As Leibniz writes in the same *Letter*: «J'avois considéré cette matière avant le livre de Mr. Wilkins, quand j'estois un jeune homme de 19 ans, dans mon petit livre de *Arte Combinatoria*, et mon opinion est que les Caracteres veritablement reels et philosophiques doivent repondre à l'Analyse des pensées» (*GP 3: 216*). This is in fact the most difficult but also the most fundamental task. Only the analysis of the diverse conceptual contents and of their relationships can constitute the limited set of primitive terms, on the basis of which the philosophical language must be constructed and which can support the choice of the

⁸ Partially reproduced in Adam-Tannery, I, Correspondance, pp. 572-3.

⁹ Of which hardly any fragment exists except some propositions in the *Traité de l'Harmonie Universelle* (1636).

¹⁰ «...for all discourses being resolved in sentences, those into words, words signifying either simple notions or being resolvable into simple notions, it is manifest that if all the sorts of simple notions be found out, and have symbols assigned to them, those will be extremely few in respect of the other», Ward (1954: 21).

¹¹ For a detailed description of Dalgarno's philosophical language, cf. Couturat (1901: 544-548), Rossi (1960: 218 segs.) and Cram (1980).

¹² Wilkins had the collaboration of John Ray, Francis Willoughby and Samuel Pepys.

¹³ For a detailed description Wilkins project, cf. Couturat, «*Sur la langue philosophique de Wilkins*» (1901: 548-552).

non-arbitrary system of signs for the philosophical language. As Leibniz says: «the name which, in this language, should be attributed to gold will be the key of the knowledge which could be obtained from it.»¹⁴

Leibniz was aware of all of his predecessors and contemporary proposals for an universal language, both international and philosophical. He fully realized both the scientific and the logical character of those projects. He was the one who have articulated more deeply the logical, semantic and heuristic objectives of such an endeavour. Against Bacon, he stressed the need of beginning the construction of universal language by the attribution of a character, not to each one of possible ideas as Bacon defended, but to each of the elementary ideas to which all others could be reduced. Against Descartes who finally condemned the project because, in his view, only the complete (encyclopaedic) analysis of the diverse conceptual contents and of their relationships might allow the finding of the primitive terms on the basis of which the philosophical language could be constructed,¹⁵ Leibniz defended that the two tasks - *characteristica universalis* and *encyclopaedia* – could be done in a zig-zag process, not as linear and sequential enterprises but as reciprocal and mutual conditioning endeavours:¹⁶ each time analysis advances and is able to determine a new unit, this unit will only become clear and distinct for though when it is designated by a specific sign. On the other hand, from the attribution of this sign will arise new heuristic virtualities as a result of its insertion into the structural network of all previously constituted signs.

Further, Leibniz was quite conscious that, in order to construct such a well grounded *organon* of rational inquiry, a new conception of symbolism and a new conception of reason should be worked out. The first, which I cannot present here¹⁷, was for the most part constructed against Descartes, in the context of their different understanding of the success of mathematical sciences and of the gnoseological status of its symbolism. The second comes mainly from Hobbes.

3. Hobbes and Leibniz

In *Leviathan*, Hobbes states: “reasoning is nothing but reckoning”¹⁸. And he adds : “when a man Reasoneth, hee does nothing else but conceive a summe totall,

¹⁴ «Nomen tamen quod in hac lingua imponetur, clavis erit eorum omnium quae de auro humanitus, id est ratione at que ordine scribi possunt, cum ex eo etiam illud appariturum sit, quatenam experimenta de eo cum ratione institui debeant» (GP 7: 13).

¹⁵ As Descartes stated in the referred *Letter to Mersenne, 20 th November 1629* «L'invention de cette langue dépend de la vrai philosophie; car il est impossible autrement de dénombrer toutes les pensées des hommes, et de les mettre par ordre, ni seulement de les distinguer en sorte qu'elles soient claires et simples» (AT1: 82).

¹⁶ In fact, in the mole of Leibniz manuscripts, there is a copy of the letter from Descartes to Mersenne in which, by his own hand, Leibniz wrote: “Il est vray que ces caractères pressupposeroient la veritable philosophie (...) Cependant, quoique cette langue depende de la vraye philosophie, elle ne depend pas de sa perfection. C'est à dire, cette langue peut estre établie, quoyque la philosophie ne soit pas parfaite: et à mesure que la science des hommes croistra, cette langue croistra aussi.» (C: 28)

¹⁷ For a presentation of this debate, cf. Pombo (1990).

¹⁸ Hobbes (*L*: 99).

from Addition of parcels; or conceive a Remainder, from Substraction of one summe from another” (L: 110). Further he explains: “those operations are not incident to Numbers onely, but to all manner of things that can be added together and taken one of another” (L: 110), that is, to arithmetic, geometry, logic, and also politics, law, “In summe, in what matter so ever there is place for addition and substaction, there also is place for Reason, and where these have no place, there Reason has nothing at all to do» (L: 110-1). Of course, the aim of extending the same kind of rational procedures to the totality of the domains finds an immediate echo in Leibniz. However, if Hobbes is the indisputable predecessor of the computational linguistic theory of cognition, there is not in Hobbes any methodological solution for the effective realization of such a rational enterprise. And given the poor mathematical capacity of Hobbes,¹⁹ his definition of reason is probably not much more than a empty *dictum*, whose rich and profound consequences Hobbes himself was not fully aware. As Couturat shows (1901: 461), Hobbes would not have penetrated all the meaning of his own formula. By contrast, Leibniz was completely prepared to develop that computational conception of reason in the context of his acute mathematical capacity and his huge interest in combinatorial processes.

But there is a further aspect in which Hobbes is an important root for Leibniz computational conception of reason: Hobbes’s conception of reason as a linguistic activity, a faculty which only operates on the basis of names.²⁰ According to Hobbes, only language makes us able to “transfer (*translation*) our mental discourse into verbal discourse” (L. p. 100), that is, only language stabilizes the fluid of mental discourse by establishing points of orientation around which representations become fixed and isolated. As Hobbes writes: “it is by the very names that we are able to stabilize the multiple conceptions of one representation” (H. N. V, § 4). That is to say, only language provides the symbolic elements upon which the activity of calculus may be realised: “For Reason, in this sense, is nothing but Reckoning (that is Adding and Substracting) of the Consequences of generall *names* agreed upon, for the marking and signifying of our thoughts” (L: 111, our emphasis). That means that without language there will be no reason. That means that ideas are not images but words.

We are here face to a very important contribution of Hobbes to Leibniz’s philosophy of language and to the computational conception of reason in general. If language is the verbal *translatio* of a previous *Mentall Discourse*²¹, that imagnetic, unstable fluid of thoughts²² in which language alone may establish points of reference, areas of security, marks, notes, names, therefore language is the sensible support for thought providing the material, signifying conditions required for the development of calculation. Only language permits the substitution of calculation of ideas by calculation of names. Only language preserves the possibility of recollecting

¹⁹ For an analysis of the polemics between Hobbes and some great mathematicians of his time, like Wallis, Ward, Boyle and Huygens, cf. Breidert (1979).

²⁰ As Hobbes states: “Children therefore are not endued with Reason at all, till they have attained the use of Speech» (L: 116).

²¹ «The general use of Speech is to transferre our Mentall Discourse, into verbal» (*Leviathan*: 100).

²² «By consequence, or Trayne of Thoughts I understand that sucession of one thought to another which is called (...) Mentall Discourse» (L: 94).

previous steps, of revising deductive chains, of progressing, gradually and securely, from one consequence to another. That is why Hobbes writes: "The use and End of reason, is not the finding of the *summe*, and truth of one, or a few consequences, remote from the first definitions, and settled significations of names; but to begin at these; and proceed from one consequence to another» (L: 112).

Hobbes is here giving a significant contribution to Leibniz who will fully adopt this thesis. And indeed, Leibniz recognizes his heritage from Hobbes precisely in this point. As he says: "names are not only signs of my present thoughts for the others but notes of my previous thoughts to myself, as Thomas Hobbes has demonstrated."²³

From this important thesis Leibniz will work out epistemic (heuristic) consequences which Hobbes never suspected. But for doing this, Leibniz will have to overcome the internal limits of Hobbes philosophy of language. Because Hobbes is attached to an evocative conception of the sign, because he claims that names must always be accompanied by the conceptions they mean, Hobbes will refuse vehemently the hypothesis of a manipulation of signs. Without the evocative fulfilment of signs by the ideas that correspondent to them, *ratio*, says Hobbes, would be reduced to *oratio* (cf. HN, V, § 14), to mechanical speech, similar to what happens with "beggars, when they say their paternoster putting together much words, and in such manner, as in their education they have learned from their nurses, from their companions, or from their teachers, having no images or conceptions in their minds answering to the words they speak» (HN, V, § 14).

Now, on the contrary, to wish for constantly thinking the meaning of signs which one handles is, not only *de facto* an impossible activity, as it would be *de jure* an obstruction to rigour and invention which both claim for the abandon of the spirit to the formal mechanisms created by the spirit. That is the main point of the celebrated leibnizian theory of *cogitation caecae*.²⁴ But that will be one of the greatest discoveries from Leibniz's philosophy of language.

4. Concluding remarks

Leibniz contribution to the computational conception of reason comes thus from three important roots which Leibniz was able to integrate and simultaneously to overcome. 1) Leibniz accepts Lull's idea of a combinatory however being able to substitute the mechanical combinatorial procedures proposed by Lull by calculatory processes of mathematical analysis (in this point anticipating Boole); 2) in the line of Bacon, Leibniz takes the communicative aims of all projects for universal languages however trying to give them a cognitive foundation in order to make them true philosophical languages. In order to do so, Leibniz agrees to the methodological regime proposed by Descartes for the construction of a philosophical language however refuting the linear relationship Descartes established between philosophical language and encyclopaedia (against the separation which later Frege will establish

²³«Verba enim non tantum signa sunt cogitationis meae praesentis ad alios, sed et notae cogitationes meae praeteritae ad me ipsum, ut demonstravit Thomas Hobbes», (A. VI, 1. 278).

²⁴ Which we have studied in Pombo (1998)

between those two issues²⁵), 3) Leibniz works out the computational conception of reason formulated by Thomas Hobbes however building a new conception of sign which makes possible to explore its heuristic potentialities.

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²⁵ Apart from the similarities between the leibnizian *characteristica universalis* and Frege's ideographic project, the two projects should be fundamentally distinguished from one another by the fact that Frege, leaving to the sciences the labour of defining their own concepts, rejects the close articulation established by Leibniz between characteristic language and encyclopaedia.

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